

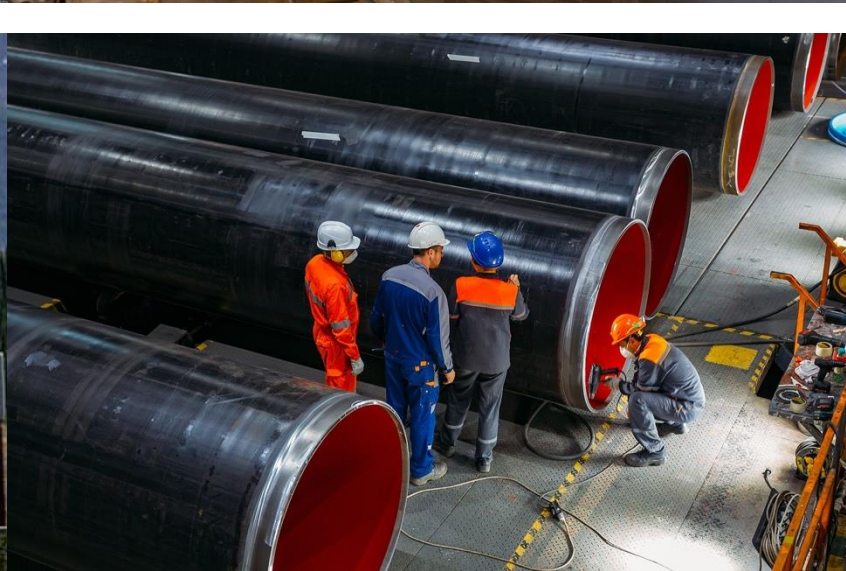
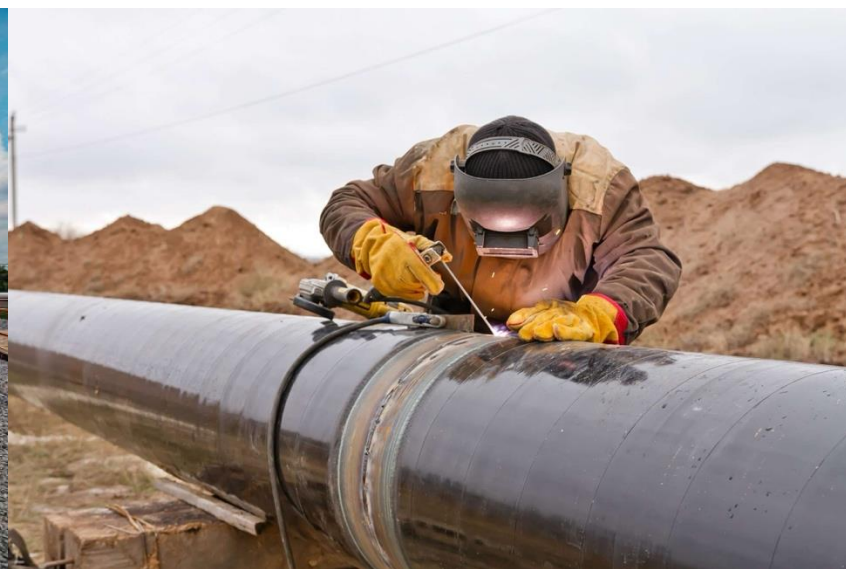
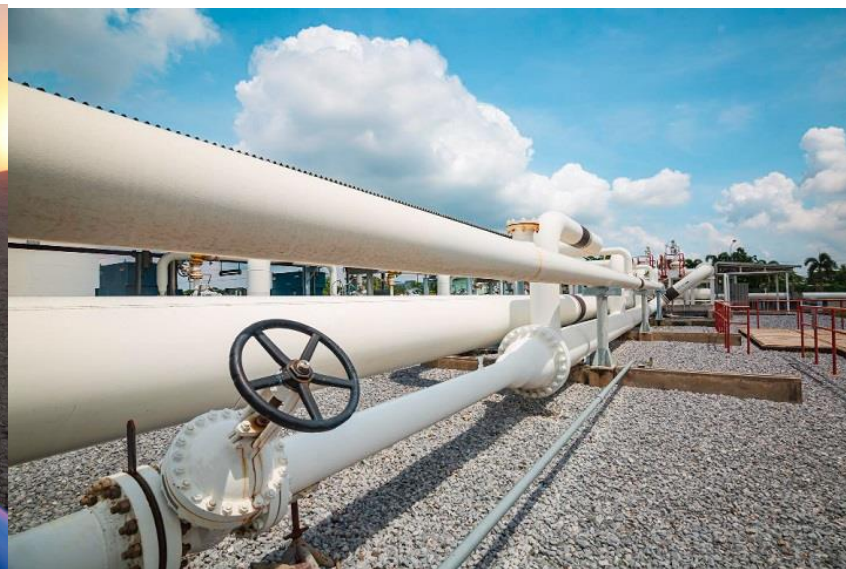
IT3212: Data Driven Software

Introduction to Pipelines

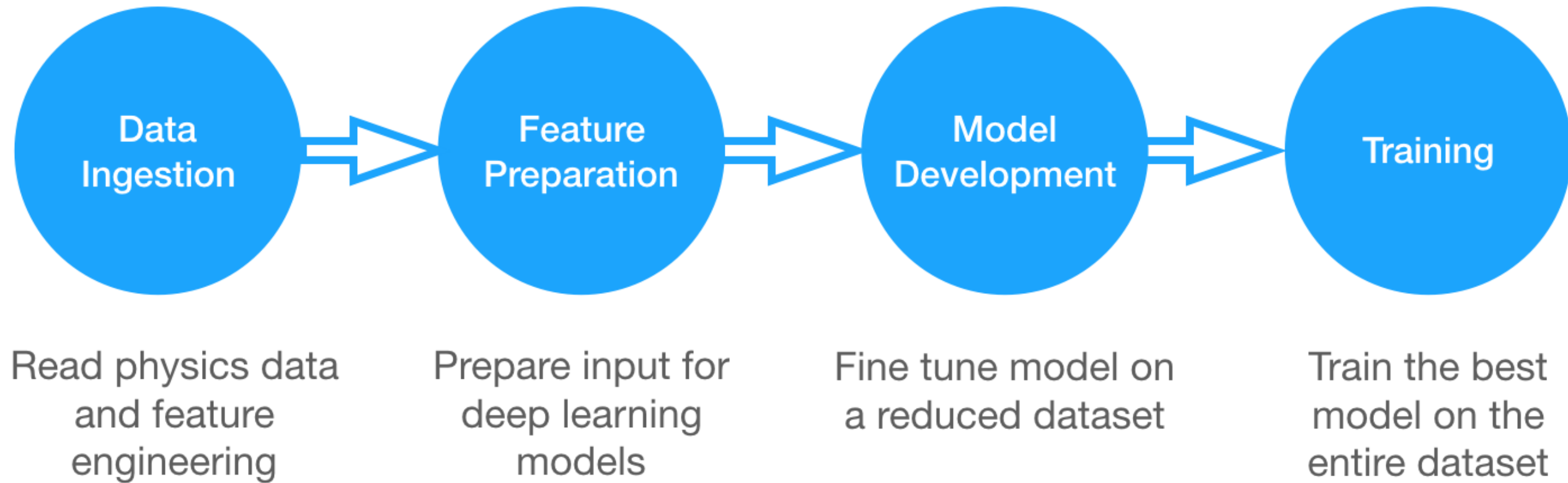
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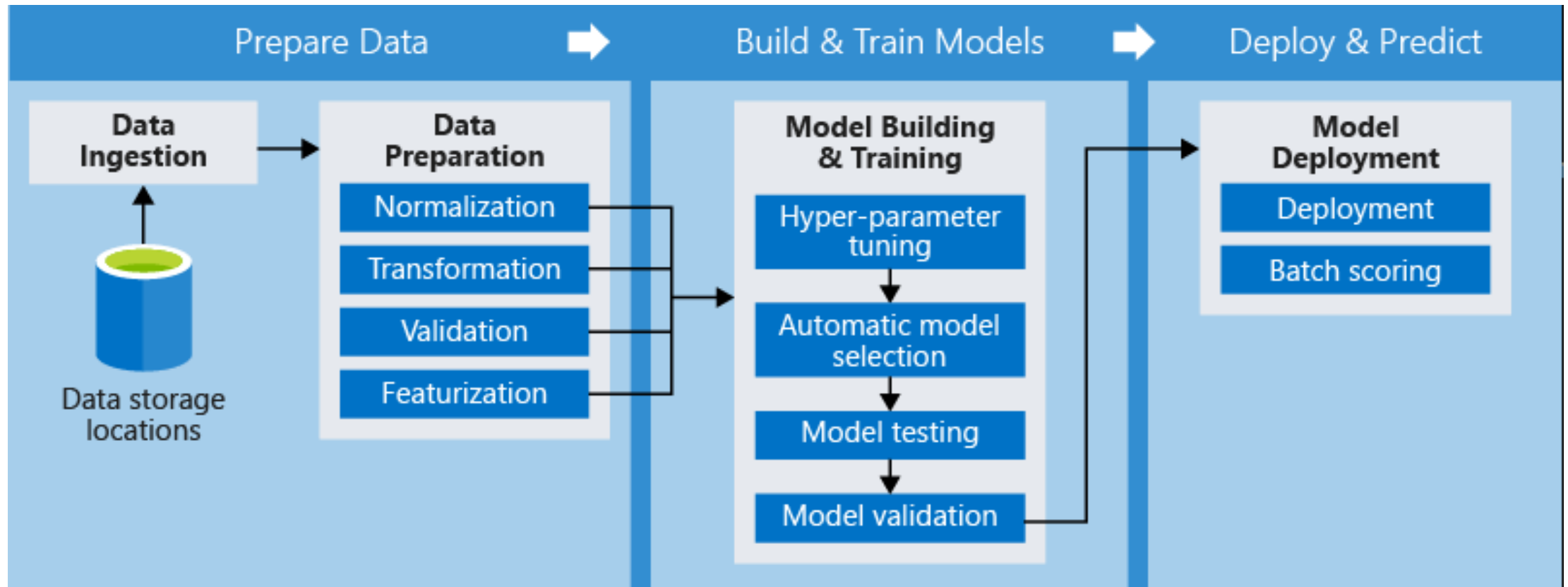
Pipelines in the real world



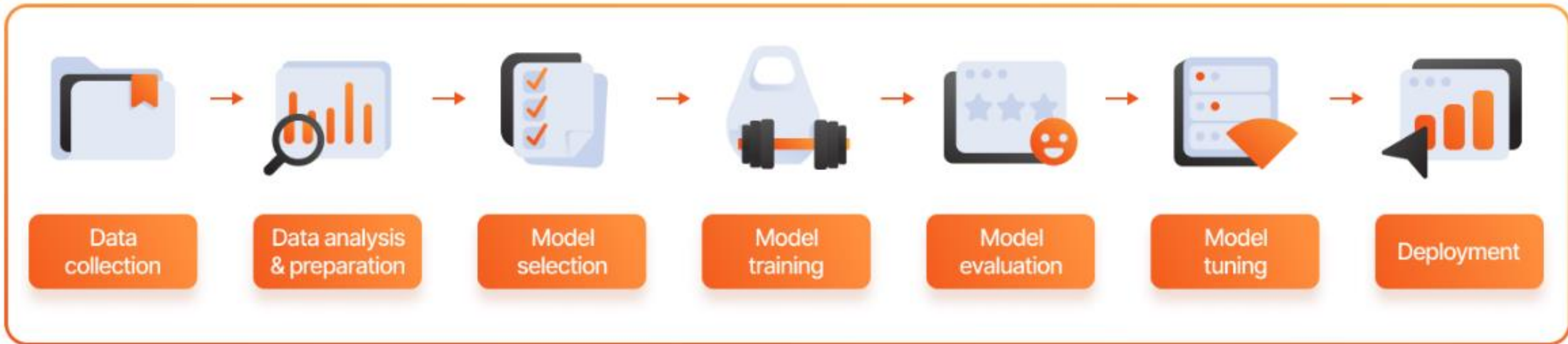
Pipeline in DDS world



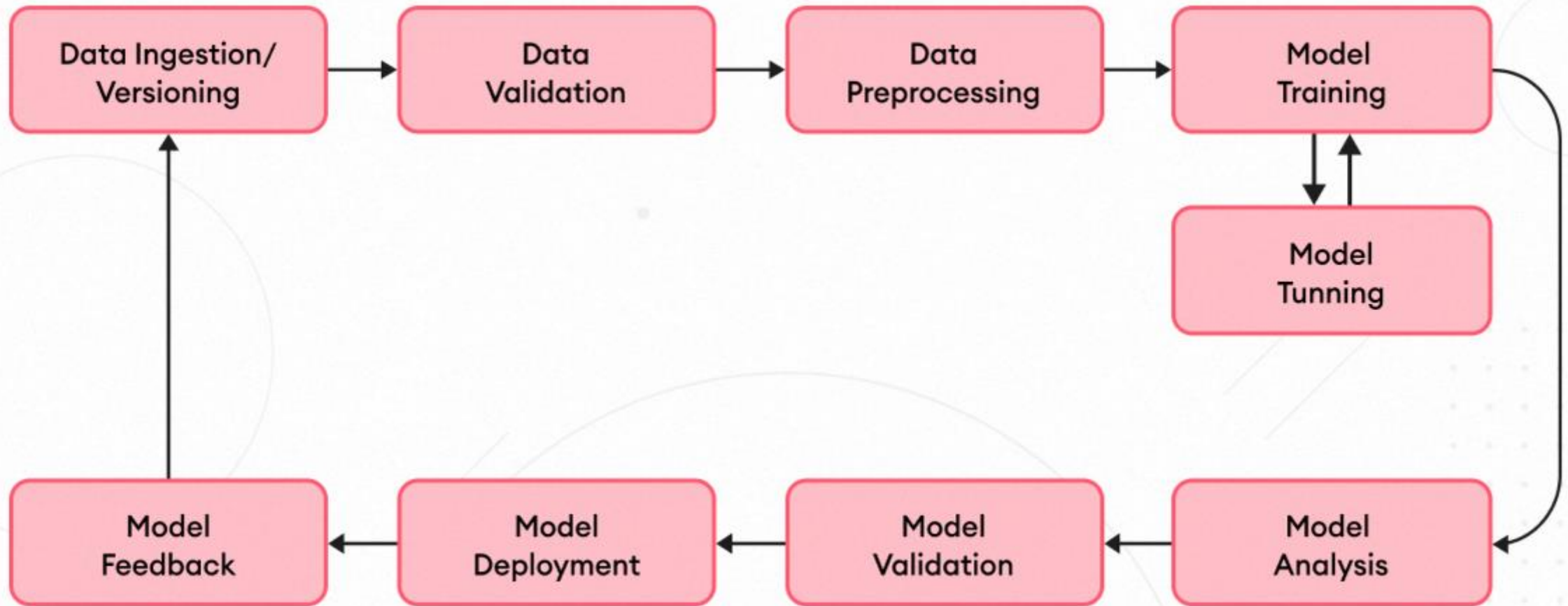
Pipeline in DDS world



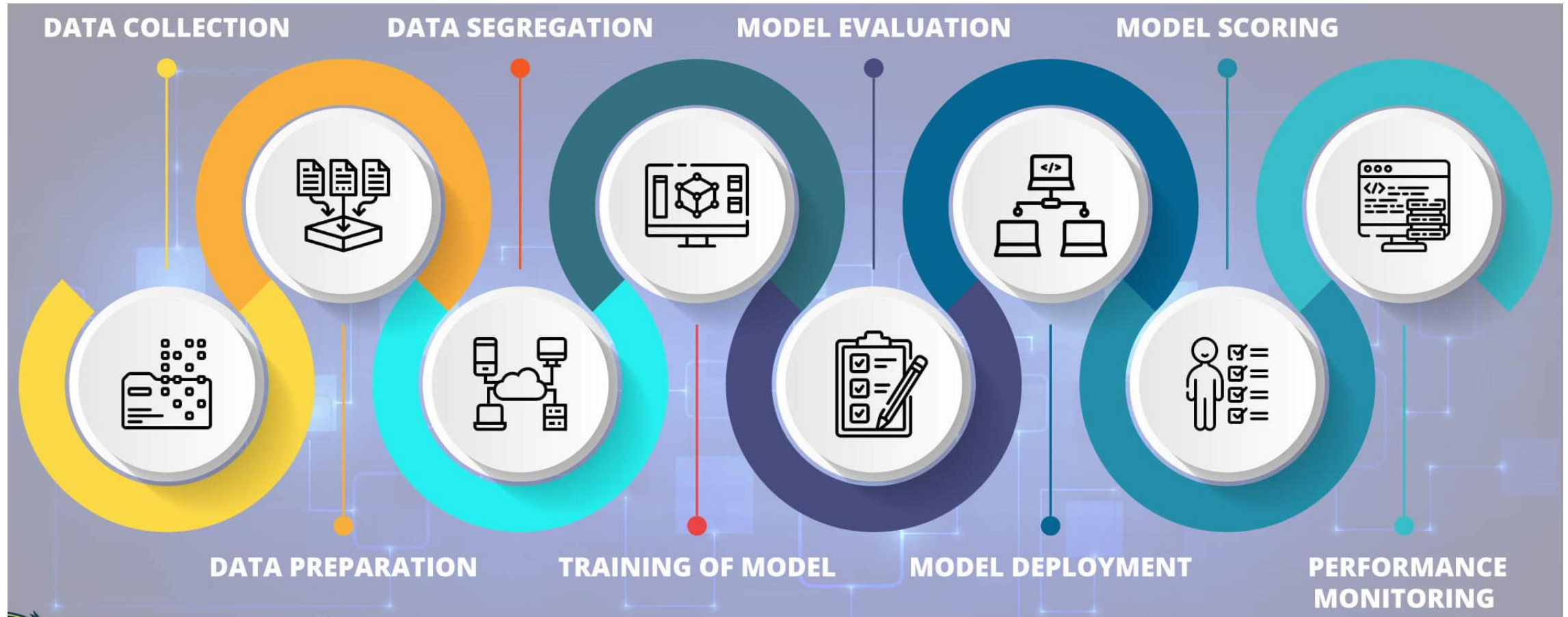
Pipelines in DDS world



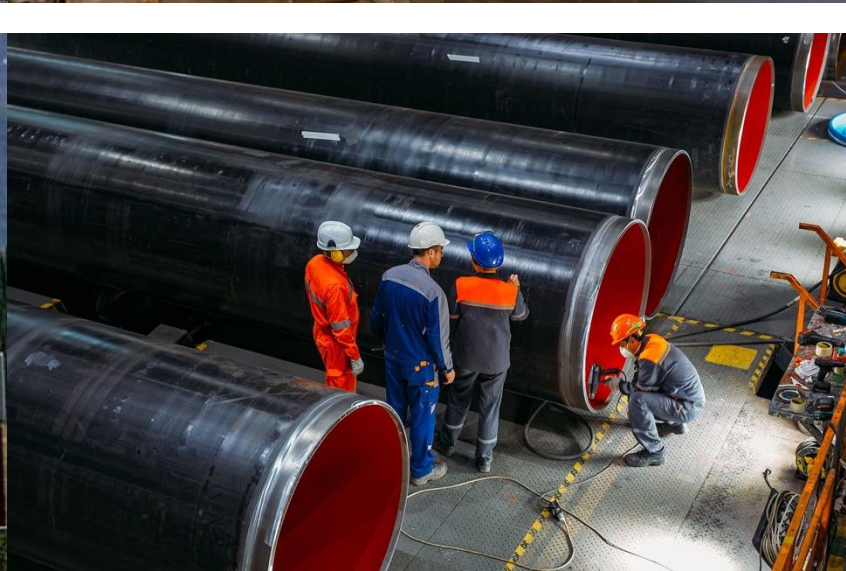
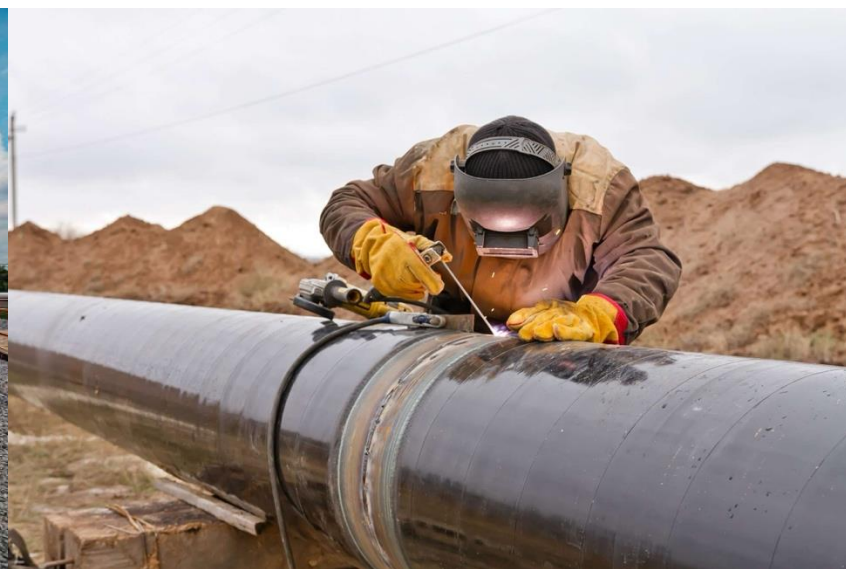
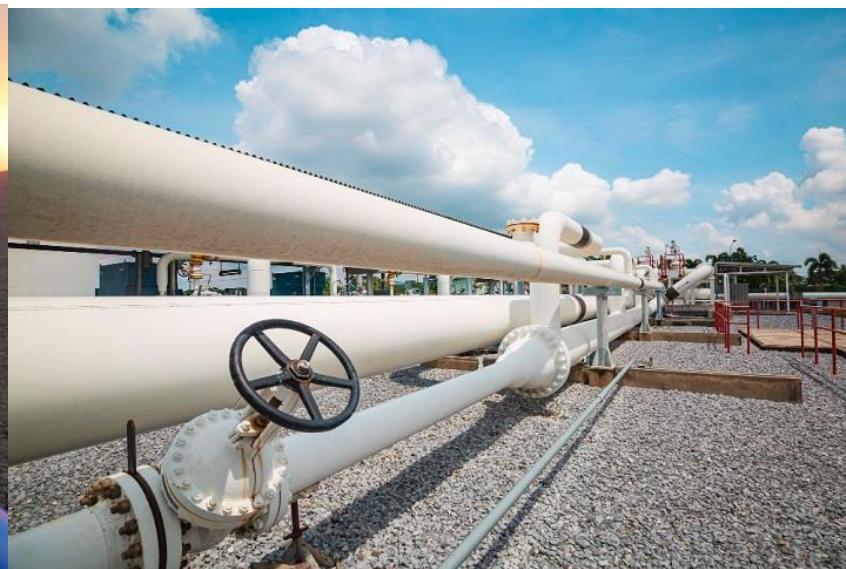
Pipelines in DDS world



Pipelines in DDS world



Pipelines in the real world



Summary of the Components of DDS pipeline

- Data collection – Data ingestion – Data storage
- Data pre-processing – Data Preparation
- Feature Preparation – Feature engineering – Feature space
- Model Training – Model Tuning – Model Selection
- Model Evaluation
- Model Deployment
- (optional) Model feedback

Data Driven Decision Making (D3M)



"I'm a software engineer, so I can confirm it works by magic."

Software Engineering



Human-Computer
Interaction



"THANKS FOR YOUR OFFER, BUT IN OUR COMPANY WE STILL HAVE A PERFECT INFORMATION AND COMMUNICATION SYSTEM. IT'S CALLED WATER COOLER TALK".

Information Systems

D3M in Software Engineering

- **Requirements Gathering and Prioritization**: Collect data on user needs, preferences, and pain points. Analyze this data to prioritize features and functionalities that will significantly impact users and align with business goals.
- **Performance Monitoring**: Continuously monitor the performance of the software, including metrics such as response times, error rates, and resource utilization. Data can help identify performance bottlenecks and guide optimizations.
- **Testing and Quality Assurance**: Use data to track the results of testing efforts, including automated tests and user acceptance testing. Analyze defect rates, test coverage, and regression test results to improve the quality of the software.

D3M in Software Engineering

- **Estimation and Planning**: Use historical project data to improve the accuracy of estimation and project planning. Data can help predict potential delays, resource requirements, and overall project timelines.
- **Code Review and Maintenance**: Analyze code quality metrics and review data to identify areas of the codebase that need improvement. This can help maintain a healthy and maintainable codebase.
- **User Behavior Analysis**: Monitor user interactions within the software to gain insights into how users engage with the product. This can guide UI/UX improvements and inform future feature development.

D3M in Software Engineering

- **A/B Testing**: Conduct A/B tests to compare different variations of a feature or UI element to determine which performs better based on user engagement and conversion metrics.
- **Feedback Analysis**: Collect and analyze user feedback and reviews to identify common issues, pain points, and areas for improvement. This data can help prioritize enhancements and bug fixes.
- **Project Retrospectives**: After each project phase or iteration, analyze data on the project's performance, team velocity, and deliverable quality. Use this data to improve processes and identify areas for continuous improvement.

D3M in Software Engineering

- **Resource Allocation**: Monitor resource utilization across different teams and projects. Data can help optimize resource allocation and prevent overloading specific team members.
- **Customer Satisfaction**: Collect customer satisfaction, engagement, and retention data. This information can guide decisions on product improvements and customer support strategies.

D3M in Software Engineering

- **Risk Management**: Utilize historical data on project risks and issues to better anticipate potential challenges and mitigate them proactively.
- **Decision Validation**: Use data to validate the outcomes of key decisions. If a decision was made based on certain assumptions, data can help determine whether those assumptions were correct.

D3M in Information Systems

- **Objective Insights**: Data-driven decisions are less influenced by biases and subjectivity, leading to more objective and fact-based choices.
- **Improved Accuracy**: Analyzing large volumes of data can lead to more accurate predictions and better understanding of complex situations.
- **Identifying Trends**: Data analysis helps in identifying trends, patterns, and outliers that might not be apparent through manual analysis.
- **Optimized Processes**: Organizations can optimize their operations, resources, and processes based on data insights, leading to increased efficiency and cost savings.

D3M in Information Systems

- **Enhanced Innovation**: Data-driven insights can uncover new opportunities for innovation and growth that might have been overlooked otherwise.
- **Quick Responses**: Organizations can respond more quickly to market or business environment changes by using real-time data to make timely decisions.
- **Competitive Advantage**: Organizations that effectively use data-driven decision-making can gain a competitive edge by making more informed choices compared to their competitors.
- **Risk Management**: Data analysis helps in identifying potential risks and vulnerabilities, allowing organizations to proactively address them.

D3M in Human-Computer Interaction

- **User Research**: Conducting user research to understand user needs, behaviors, and preferences is fundamental. This may involve methods such as surveys, interviews, usability testing, and ethnographic studies. The data collected helps designers and developers gain insights into user motivations, pain points, and expectations.
- **Quantitative Data**: Quantitative data provides measurable metrics, such as user engagement, click-through rates, completion rates, and time spent on specific tasks. This data is often collected through analytics tools embedded within interactive systems or websites. Quantitative data helps identify patterns and trends in user behavior.

D3M in Human-Computer Interaction

- **A/B Testing**: A/B testing involves comparing two or more variations of a design to determine which performs better in terms of user engagement, conversion rates, or other relevant metrics. Researchers can identify which changes lead to improved outcomes by randomly assigning users to different design versions.
- **User Behavior Analysis**: Analyzing user interactions within a system provides insights into navigation patterns, feature usage, and areas where users might face difficulties. Heatmaps, session recordings, and clickstream analysis can help uncover specific pain points.

D3M in Human-Computer Interaction

- **Persona Development**: Personas are fictional representations of different user types based on real data. These personas guide design decisions by clearly understanding user characteristics, goals, and preferences.
- **Iterative Design**: Data-driven decision-making supports an iterative design process. Designers use collected data to refine and optimize designs, gradually improving user experiences over time.
- **Accessibility Insights**: Data-driven approaches can help identify accessibility issues by analyzing user interactions. This is crucial for ensuring that interactive systems are usable by diverse users.

D3M in Human-Computer Interaction

- **Predictive Modeling**: Machine learning and predictive modeling can be used to anticipate user behavior and adapt interfaces accordingly. For instance, recommending content or features based on user history and preferences.
- **Ethical Considerations**: Data-driven decision-making in HCI should prioritize user privacy and ethical considerations. Proper anonymization and consent mechanisms are essential when collecting and using user data.
- **Feedback Loops**: Establishing feedback loops with users allows for continuous improvement based on real-time user input and reactions.

Examples of D3M in real world

- **E-commerce: Personalized Recommendations**: E-commerce platforms use data on user browsing and purchase history to provide personalized product recommendations. This increases customer engagement, conversion rates, and overall satisfaction by tailoring the shopping experience to individual preferences.
- **Healthcare: Predictive Analytics**: Hospitals and healthcare providers use data-driven predictive analytics to forecast patient admissions and allocate resources accordingly. This improves patient care by ensuring the availability of necessary staff, equipment, and facilities.
- **Manufacturing: Supply Chain Optimization**: Manufacturing companies analyze supply chain data to optimize inventory levels, reduce lead times, and minimize production bottlenecks. This data-driven approach leads to more efficient production processes and cost savings.

Examples of D3M in real world

- **Financial Services: Fraud Detection**: Banks and credit card companies use data analytics to detect fraudulent transactions. By analyzing transaction patterns and anomalies, they can identify potentially fraudulent activities and take preventive measures.
- **Transportation: Traffic Management**: Data from traffic sensors, GPS devices, and mobile apps are used to manage urban traffic flow. City planners can optimize traffic signal timings and reroute traffic based on real-time data, reducing congestion and improving commute times.

Examples of D3M in real world

- **Education: Adaptive Learning**: Educational technology platforms use data from student interactions to provide adaptive learning experiences. Algorithms analyze student performance data to tailor lessons and assignments, ensuring that students receive targeted support and challenges.
- **Energy: Smart Grid Management**: Energy companies utilize data from smart meters and sensors to optimize energy distribution. This enables better load balancing, reduces energy waste, and enhances the electrical grid's stability.
- **Urban Planning: Public Services Allocation**: City governments use data to allocate resources for public services such as waste collection and emergency response. Analyzing data on population density and usage patterns ensures efficient service delivery.

Examples of D3M in real world

- **Agriculture: Precision Farming**: Farmers collect data from sensors and drones to monitor soil moisture, nutrient levels, and crop health. This information guides decisions about irrigation, fertilization, and pest control, leading to increased crop yields and resource efficiency.
- **Software Development: User Experience Enhancement**: Software development teams analyze user feedback and behavior to identify areas for improvement. Data-driven decisions can lead to user interface adjustments, bug fixes, and feature enhancements that align with user needs.

Examples of D3M in real world (IS)

- **Netflix: Content Recommendations**: Netflix uses data from user viewing history, ratings, and interactions to recommend personalized content to its users. This data-driven approach has contributed significantly to user engagement and retention.
- **Amazon: Inventory Management**: Amazon's fulfillment centers rely on data-driven algorithms to manage inventory levels. The system predicts demand based on historical sales data, allowing Amazon to optimize storage space, reduce excess inventory, and maintain efficient order fulfillment.

Examples of D3M in real world (IS)

- **NASA: Space Exploration**: NASA collects and analyzes vast amounts of data from spacecraft, rovers, and telescopes to inform space exploration decisions. Data-driven insights have contributed to discoveries about Mars' geology, the composition of distant planets, and more.
- **Walmart: Supply Chain Optimization**: Walmart uses data analytics to optimize its supply chain by predicting demand, tracking inventory levels, and streamlining logistics operations. This has led to reduced stockouts and improved product availability.
- **Airbnb: Dynamic Pricing**: Airbnb employs data-driven dynamic pricing to adjust rental rates based on factors such as location, demand, seasonality, and local events. This approach maximizes revenue for hosts and provides competitive pricing for guests.

Examples of D3M in real world (IS)

- **Procter & Gamble: Product Innovation**: Procter & Gamble uses data-driven insights from consumer feedback and market trends to inform product development and innovation. This approach has led to the creation of new products that better align with customer preferences.
- **Facebook: News Feed Customization**: Facebook's News Feed algorithm uses data from user interactions to customize the content users see on their feeds. This data-driven approach aims to show users the most relevant posts and advertisements.
- **City of Chicago: Predictive Policing**: The Chicago Police Department uses data-driven predictive policing to allocate resources and focus law enforcement efforts on areas with higher likelihoods of criminal activity. This approach helps prevent crime and enhance public safety.

Examples of D3M in real world (SE)

- **Google Search Algorithm**: Google uses a complex data-driven algorithm to rank search results based on relevance and user behavior. The algorithm considers factors like user click-through rates, bounce rates, and site quality to continuously improve search results.
- **Adobe Creative Cloud**: Adobe collects data on how users interact with its Creative Cloud suite, including which tools are used most frequently and which features are rarely utilized. This data informs decisions about feature updates and improvements to enhance user productivity.
- **Microsoft Office 365**: Microsoft uses data from Office 365 users to identify common pain points and areas where users might struggle with specific tasks. This data helps shape user interface changes, feature additions, and overall user experience enhancements.

Examples of D3M in real world (SE)

- **Salesforce: Sales Predictions**: Salesforce uses data analytics to predict sales trends and customer behavior. This helps sales teams prioritize leads, tailor communication strategies, and make more accurate forecasts.
- **Atlassian Jira: Agile Project Management**: Atlassian's Jira platform collects data on task progress, team velocity, and issue resolution times. These data points inform agile project management decisions like sprint planning and backlog prioritization.
- **GitHub: Code Repository Insights**: GitHub provides data-driven insights to developers about their code repositories. This includes data on pull request acceptance rates, code review turnaround times, and contribution statistics, enabling developers to make informed decisions about project management and collaboration.

Examples of D3M in real world (SE)

- **Bug Tracking and Issue Management: Jira Software**: Development teams use Jira Software to track bugs, issues, and feature requests. Data from these platforms helps prioritize bug fixes, allocate resources effectively, and improve software quality over time.
- **App Store Optimization**: Developers and marketers use data-driven approaches to optimize app store listings. This includes analyzing keywords, monitoring user reviews, and tracking download and retention rates to improve app visibility and user acquisition.
- **Website Analytics: Google Analytics**: Google Analytics provides data-driven insights into website traffic, user behavior, and conversion rates. Businesses use this data to refine their websites, identify areas for improvement, and optimize user journeys.

Examples of D3M in real world (HCI)

Netflix's User Interface Design: Netflix uses data analytics to understand how users interact with its platform. This data informs decisions about the placement of content, the design of recommendation algorithms, and the development of features to enhance user engagement.

Website Heatmaps and Interaction Analysis: HCI researchers use tools like Crazy Egg or Hotjar to generate heatmaps that visualize where users click, scroll, and interact most on a website. This data helps designers optimize layouts, call-to-action placements and content organization.

Mobile App Analytics: Mobile app developers use tools like Mixpanel or Firebase to track user interactions within their apps. By analyzing data on user flows, drop-off points, and feature usage, they can identify areas for improvement and prioritize updates.

Examples of D3M in real world (HCI)

Gaming: Player Behavior Analysis: In the gaming industry, data-driven analysis of player behavior helps improve game design. By tracking in-game actions, interactions, and preferences, game developers can adjust gameplay mechanics and create more engaging experiences.

User Testing and Eye Tracking: Researchers use eye-tracking technology to understand where users look and focus their attention on a screen. This data aids in optimizing user interfaces, ensuring that key elements are noticed and making navigation more intuitive.

Social Media Platform Design: Social media platforms like Facebook and Twitter analyze user engagement metrics, such as likes, shares, and comments, to optimize the design of their feeds and algorithms. This ensures users are exposed to content relevant to their interests.

Examples of D3M in real world (HCI)

Accessibility and User Need: Data-driven insights about user accessibility needs influence the design of software and interfaces to ensure they are usable by people with disabilities. For example, screen reader usage data helps developers create more accessible web content.

Cars: Driver Behavior Analysis: Car manufacturers analyze data on driver behavior, such as how often certain controls are used and driver distraction levels, to design safer and more intuitive user interfaces for in-car systems.

Examples of D3M in real world (HCI)

Healthcare Technology: Patient Interaction Analysis: In healthcare, user interactions with medical devices and electronic health records are analyzed to design more user-friendly interfaces. This includes interfaces for doctors, nurses, and patients.

E-Learning Platforms: User Progress Tracking: E-learning platforms like Coursera or Udemy track user progress, quiz results, and engagement metrics to personalize learning experiences, offer relevant recommendations, and identify areas where learners might struggle.

Requirements for effective D3M

- Define clear objectives and key performance indicators (KPIs) for your project.
- Ensure data is collected accurately and consistently.
- Establish a culture of data-driven decision-making within your team.
- Use appropriate tools for data analysis and visualization.
- Regularly review and update your data-driven strategies based on the insights gained.

Benefits of D3M

- **Objective Insights**: Data-driven decisions are based on empirical evidence rather than personal opinions or biases. This reduces the influence of subjective judgment and leads to more objective and informed choices.
- **Accuracy and Precision**: Data collection and analysis methods aim to provide accurate and precise information. This helps in understanding patterns, trends, and correlations that might be missed through anecdotal observations.
- **Improved Decision Quality**: Data-driven decisions are grounded in real-world information, leading to better-informed choices. This, in turn, can lead to improved outcomes, whether in business strategies, product design, or policy-making.

Benefits of D3M

- **Identification of Trends and Patterns**: Analyzing large datasets allows for the identification of trends, patterns, and anomalies that can provide valuable insights for predicting future outcomes or identifying emerging opportunities or challenges.
- **Quick Adaptation**: Data-driven decision-making facilitates quick adaptation to changing circumstances. As new data becomes available, decisions can be adjusted or refined to align with the latest information.
- **Enhanced Strategic Planning**: Organizations can use data to gain a deeper understanding of market dynamics, customer preferences, and competitive landscapes. This enables more effective strategic planning and positioning.

Benefits of D3M

- **Risk Management**: Data-driven approaches can help identify potential risks and uncertainties. By analyzing historical data and trends, organizations can make informed choices to mitigate risks.
- **Efficiency and Resource Allocation**: Data-driven decisions allow for optimal allocation of resources. This is particularly valuable in fields such as supply chain management, where efficient resource distribution can lead to cost savings and improved performance.
- **Personalization and Customer Satisfaction**: In customer-centric industries, data-driven decision-making enables personalized experiences. Businesses can tailor products, services, and marketing strategies based on individual customer preferences and behavior.

Benefits of D3M

- **Innovation and Creativity**: Data-driven insights can inspire innovation by revealing untapped opportunities or areas of improvement that might not be immediately obvious. This can lead to the development of novel products, services, or solutions.
- **Measurable Outcomes**: Data-driven decisions are often accompanied by clear metrics and key performance indicators (KPIs). This allows for the measurement of outcomes and the assessment of whether decisions have achieved their intended results.
- **Collaboration and Communication**: Data-driven decision-making can foster better collaboration among teams. Data serves as a common language that facilitates communication and aligns various stakeholders around a shared understanding.

Benefits of D3M

- **Transparency and Accountability**: Data-driven decisions are traceable and accountable, as they are based on documented information. This promotes transparency within organizations and helps in justifying decisions to stakeholders.
- **Continuous Improvement**: Data-driven decision-making promotes a culture of continuous improvement. By analyzing data, organizations can identify areas that require attention and implement iterative changes for ongoing enhancement.
- **Evidence-Based Policy-Making**: Data-driven approaches are crucial in government and public policy settings. Policymakers can make informed decisions by considering data on social, economic, and environmental factors.

Cautions

- Remember that while data-driven decisions are valuable, they should be complemented by human expertise and context.
- Data is a tool to inform decisions, but human judgment and experience are crucial in interpreting and applying the insights gained from the data.
- To implement effective data-driven decision-making, organizations need to have the right technological infrastructure, data governance practices, skilled personnel (data analysts, data scientists, etc.), and a culture that values data-driven insights.

What's next?

Next Lectures

- Data preprocessing
- Basic data analysis and Visualizations

Assignment Step:

- Start thinking of your interest, project
- Start making groups

Thank you!!
Questions??